

Zechen Li

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Education

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| Ph.D. | University of New South Wales , Sydney, Australia
Computer Science Supervisor: Prof. Flora Salim, Dr. Hao Xue | 2023 – 2027 (expected) |
| M.S. | Northeastern University , Boston, USA
Data Analytics Engineering GPA: 3.917/4.0 | 2019 – 2021 |
| B.Eng. | Southern Medical University , Guangzhou, China
Biomedical Engineering Minor: Information Technology GPA: 3.09/4.0 | 2015 – 2019 |

Research Interests

- Multimodal learning, wearable sensor–language alignment, and LLMs for health and human activity recognition.
- Advancing verifiable and structured reasoning on wearable health data through domain knowledge integration and evidence retrieval with LLMs.

Selected Publications

(*: equal contribution; [†]: corresponding author)

- **Zechen Li**[†], Keerthana Natarajan, Weizhi Zhang, Menglian Zhou, Simon A. Lee, Yuwei Zhang, Maxwell A. Xu, Zeinab Esmaeilpour, Flora D. Salim, Mark Malhotra, Lindsey Sunden, Shwetak Patel, Yuzhe Yang[†], Ahmed A. Metwally[†]. *GlucFM: A Dual-Stream Foundation Model for Continuous Glucose Monitoring*. Preprint, 2026.
- Hada Melino Muhammad*, **Zechen Li***, Flora Salim, Ahmed A. Metwally. *CGM-JEPA: Learning Consistent Continuous Glucose Monitor Representations via Predictive Self-Supervised Pretraining*. Preprint, 2026.
- **Zechen Li**, Baiyu Chen, Hao Xue, Flora D. Salim. *ZARA: Training-Free Motion Time-Series Reasoning via Evidence-Grounded LLM Agents*. Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL Oral), 2026.
- **Zechen Li**, Shohreh Deldari, Linyao Chen, Hao Xue, Flora D. Salim. *SensorLLM: Aligning Large Language Models with Motion Sensors for Human Activity Recognition*. Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP Main), 2025.
- **Zechen Li**, Anders Søgaard. *QLEVR: A Diagnostic Dataset for Quantificational Language and Elementary Visual Reasoning*. Findings of the North American Chapter of the Association for Computational Linguistics (NAACL Findings), 2022.
- Meng Zhou*, **Zechen Li***, Pengtao Xie. *Self-supervised Regularization for Text Classification*. Transactions of the Association for Computational Linguistics (TACL), 2021. (Oral Presentation at NAACL 2021)
- Sparsh Rastogi, Tanmay Kumar, Baiyu Chen[†], Jatin Bedi, **Zechen Li**[†], Flora D. Salim[†]. *TRACE-TS: Attribution-Grounded and Traceable Sensor-Language Reasoning for Human Activity Understanding*. Preprint, 2026.
- Weizhi Zhang, **Zechen Li**, Hamid Palangi, Ben Graef, A. Ali Heydari, Simon A. Lee, Salman Rahman, Ray Luo, Zeinab Esmaeilpour, Erik Schenck, Chloe Zhang, Yamin Li, Menglian Zhou, Philip S. Yu, Daniel McDuff, Lindsey Sunden, Mark Malhotra, Shwetak Patel, Ahmed A. Metwally. *RubricsTree: Scalable and Evolving Open-Ended Evaluation of Personal Health Agents across Health Memory and Medical Skills*. Preprint, 2026.
- Linyao Chen, Qinlao Zhao, **Zechen Li**, Mingming Li, Likun Ni, Jinyu Chen, Yuhao Yao, Xuan Song, Noboru Koshizuka, Hiroki Kobayashi. *Towards Efficient and Evidence-Grounded Mobility Prediction with LLM-Driven Agent*. Preprint, 2026.
- Baiyu Chen, **Zechen Li**, Wilson Wongso, Lihuan Li, Xiachong Lin, Hao Xue, Benjamin Tag, Flora Salim. *AnyMo: Geometry-Aware Setup-Agnostic Modeling of Human Motion in the Wild*. Preprint, 2026.

- Ruiyi Yang, **Zeichen Li**, Hao Xue, Imran Razzak, Flora D. Salim. *MAGE: Multi-Agent Self-Evolution with Co-Evolutionary Knowledge Graphs*. Preprint, 2026.
- Baiyu Chen, Wilson Wongso, **Zeichen Li**, Yonchanok Khaokaew, Hao Xue, Flora Salim. *COMODO: Cross-modal Video-to-IMU Distillation for Efficient Egocentric Human Activity Recognition*. Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT/UbiComp), 2026.
- Meng Zhou, **Zeichen Li**, et al. *On the Generation of Medical Dialogs for COVID-19*. Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics (ACL Short), 2021.

Experience

Google, Student Researcher

Mountain View, CA
Feb 2026 – Present

- Building foundation models for continuous glucose monitoring.
- Connecting sensor foundation models with language models.

University of New South Wales, Casual Academic & Research Associate

Sydney, Australia
Aug 2022 – Dec 2025

- Taught tutorials for multiple courses:
 - COMP6713 Natural Language Processing
 - COMP9444 Neural Networks and Deep Learning
 - COMP9517 Computer Vision
 - DATA9001 Fundamentals of Data Science
- Conduct research on leveraging LLMs for wearable sensor time-series, focusing on human activity recognition and reasoning over motion sensor data.
- Mentored undergraduate students, co-supervising "Taste of Research" projects with Prof. Flora Salim.

Singapore University of Technology and Design, Visiting Research Fellow

Singapore
Mar 2023 – Jun 2023

Supervisor: Prof. Wei LU

- Conducted research on integrating human mobility data and time-series signals with language models.

University of Copenhagen, Research Assistant (part-time)

Copenhagen, Denmark
May 2021 – May 2022

Supervisor: Prof. Anders Søgaard

- Developed QLEVR, a large-scale synthetic dataset (100k images, 1M questions) for visual reasoning with quantificational language.
- Designed scene graphs and 3D image generation pipeline using Blender to support complex linguistic and spatial reasoning tasks.

University of California, San Diego, Research Assistant (part-time)

San Diego, CA
May 2020 – May 2021

Supervisor: Prof. Pengtao Xie

- Proposed and evaluated a Self-Supervised Regularization (SSL-Reg) framework for text classification using a Y-structured joint model with shared parameters.
- Developed CovidDialog (English & Chinese), one of the first large-scale medical dialog datasets for COVID-19, and explored dialog generation with self-supervised regularization.
- Demonstrated that SSL-Reg alleviates overfitting and improves generalization, especially on low-resource tasks.

Honors & Awards

- UNSW Founders New Wave Cohort 15 Showcase — Faculty of Engineering Award 2025
- UNSW Higher Degree by Research (HDR) Scholarship 2023 – Present
- The Dean's Seattle Leaders & Scholars Award, Northeastern University 2019 & 2020

- Scholarship of Academic Excellence (Top 13%), Southern Medical University 2016 & 2018
- Excellent Student Leader (Top 30%), Southern Medical University 2016 & 2018
- 3rd Prize, South China Division, WeChat Mini Program Development Competition 2018
- Outstanding Student (Top 10%), Southern Medical University 2016

Skills

- **Languages:** Chinese (native); English (proficient)
- **Programming:** Python, PyTorch, Java, R, C/C++, JavaScript, SQL